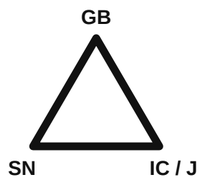
Goodman–Buckler panel

Grain free amino acid GWAS, 280 inbreds, 71 traits

Free pool dominated by Asn and Pro

How to read the gene marks



A filled corner shows which analysis found the gene.

- GB only
- SN and IC

IC or J is printed after marks at the IC / J corner.

B64 × CML14 biparental cross
Shared Asp and Pro QTL on chr 3

Other grain FAA GWAS candidates

Protein turnover and ER membrane
- Peptidase A1 (190800), 8 traits
- EMC4 (394360), rhomboid-like 19 (394380), 9 traits

N assimilation
- GOGAT (329710, 360480), GS (059870)

Transport
- NPF4.6 (277160)
- UmamiTs (379220, 207630, 352400)
- AVT6C (368680), AVT3A-like (429100)

Other soil N candidates

Within or near the 12 SNPs
- Syndetin (022740)
- RLT1 homeobox-DDT (359990)
- TTL1-like zinc finger (130210)
- Protein kinase (226000)
- HB transcription factor (265710)
- Transcription repressor (001920)

Other IC and J candidates

Protein turnover
- IC SAG39 cysteine protease (214780)
- IC ATG18a-like (352300), ATG18g (287690)
- J NPL4-like 1 (146560)

N assimilation
- IC GDH (060520, 162300), GS (162900)

Transport
- IC AMT2-like (214890), NPF3.1 (287950)
- J LHT8-like (132750), ProT-like (159580)



Aspartate-family amino acid metabolism

Shared gene, SAM synthase (354640)

Shared gene, class I Gln amidotransferase (217220)

ω-amidase 005710

Oxaloacetate

Asp aminotransferase 152450 J

Asp

Asn synthetase domain 036060, 036070

Asn

amide N

Gln

L-homoserine biosynthesis

Asp kinase 094670 J 220040

Aspartyl-4-P

Asp-semialdehyde DH 214360 IC

Asp-semialdehyde

several steps

Lys

DHDPS, DHDPR, DAPDC 389290, 176170, 406820

Homoserine DH 405410

Homoserine

Thr synthase 294790 IC

Met synthase 052750 IC

Thr

Met

SAM synthase 354640

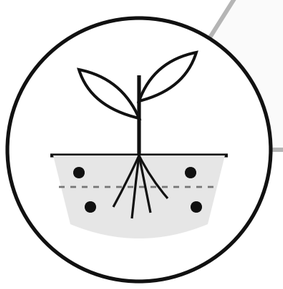
SAM

Met S-methyltransferase 356170

Ile

S-methyl-Met

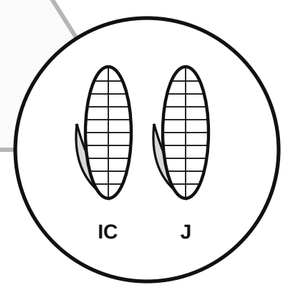
Shared gene, Met synthase (052750)



Maize landraces

Soil N GWAS, 3,107 georeferenced accessions

12 SNPs pass both specificity tests



Indian Chief and Jarvis

FST scan, Cycle 0 vs Cycle 14

IC free AA and Pro up, Asp and Glu down

J Asp fraction up (nominal only)

Genes are positional candidates, within 25 kb of a significant GWAS SNP or in a top-5% FST window. Numbers are Zm00001eb gene IDs (B73 RefGen_v5).

Shaded box, the CornCyc L-homoserine biosynthesis pathway, the only pathway with a candidate in all four gene sets and the top-ranked pathway in the recurrence test.

Thick arrows, L-homoserine biosynthesis. Dashed arrows, amide N from Gln to Asn, and the ω-amidase route returning Asn carbon to oxaloacetate. Grey, context not found in any analysis.